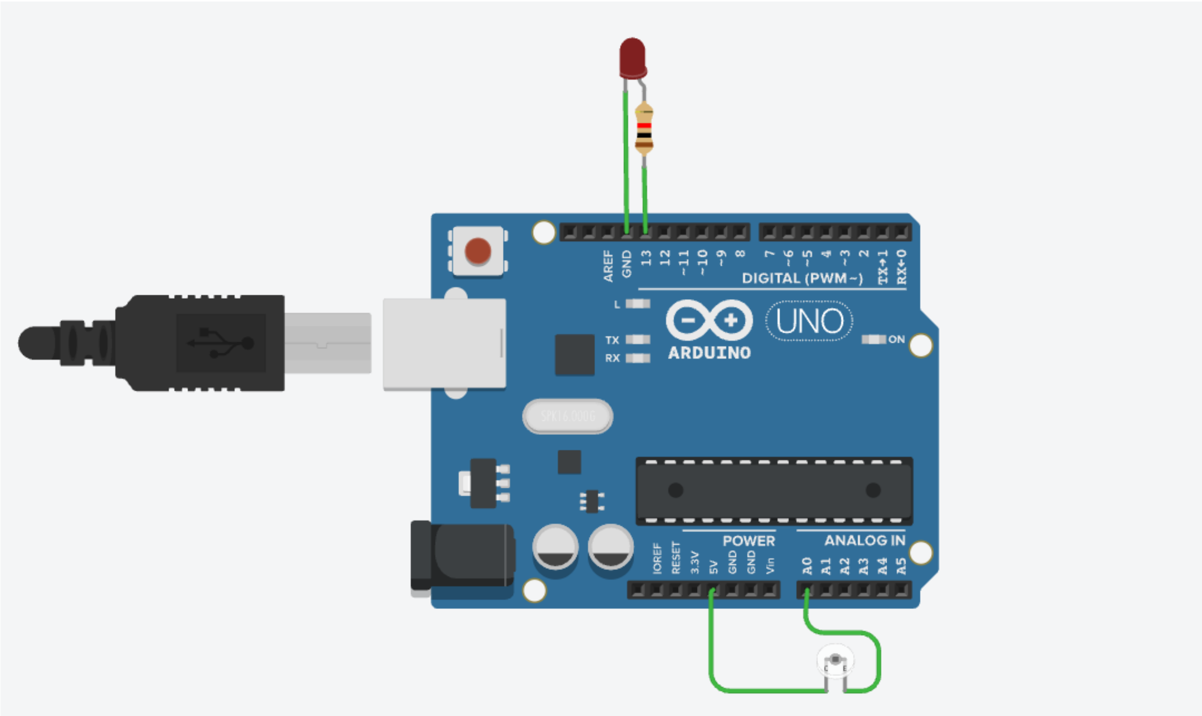
Embedded System and Internet of Things

Task:-02

1. Circuit Diagram:-



2. Source Code:-

```
int photoPin = A0;     // Phototransistor output

int ledPin = 13;       // LED pin

int threshold = 500;  // Light threshold value


void setup() {
```

```
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int lightValue = analogRead(photoPin); // Read sensor value

  Serial.print("Light Value: ");
  Serial.println(lightValue);

  if (lightValue > threshold) {     // Bright condition
    digitalWrite(ledPin, HIGH);     // LED ON
    Serial.println("Bright Light - LED ON");
  } else {                          // Dark condition
    digitalWrite(ledPin, LOW);      // LED OFF
    Serial.println("Low Light - LED OFF");
  }

  delay(500);
}
```

3. Explanation:-

A) How the Phototransistor Works:-

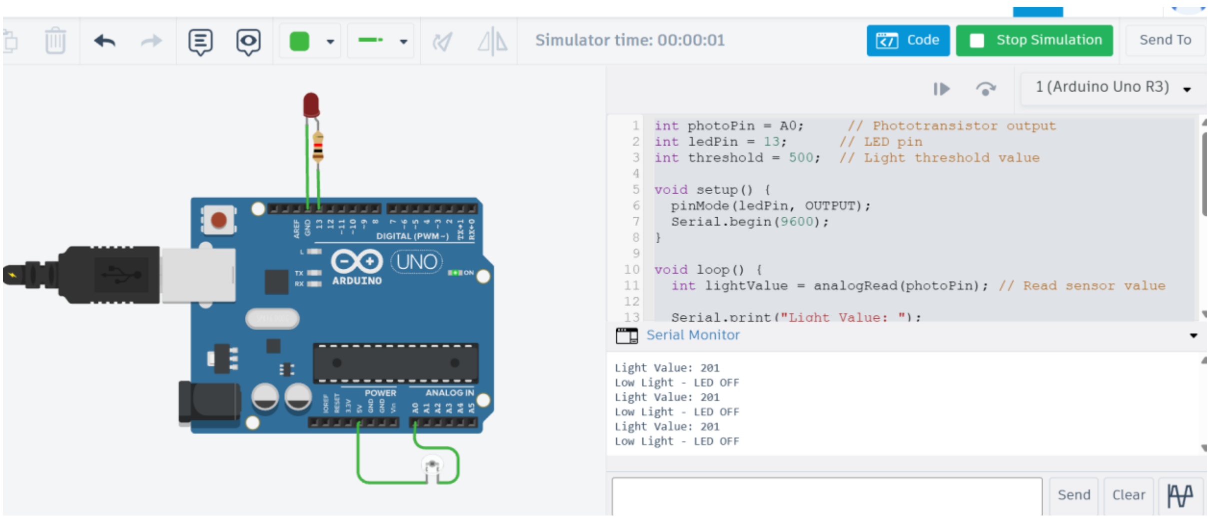
- A phototransistor changes its **current flow based on light intensity**.
- More light → more current → higher voltage at Arduino analog pin.
- Less light → lower voltage.

B) What the Code Does:-

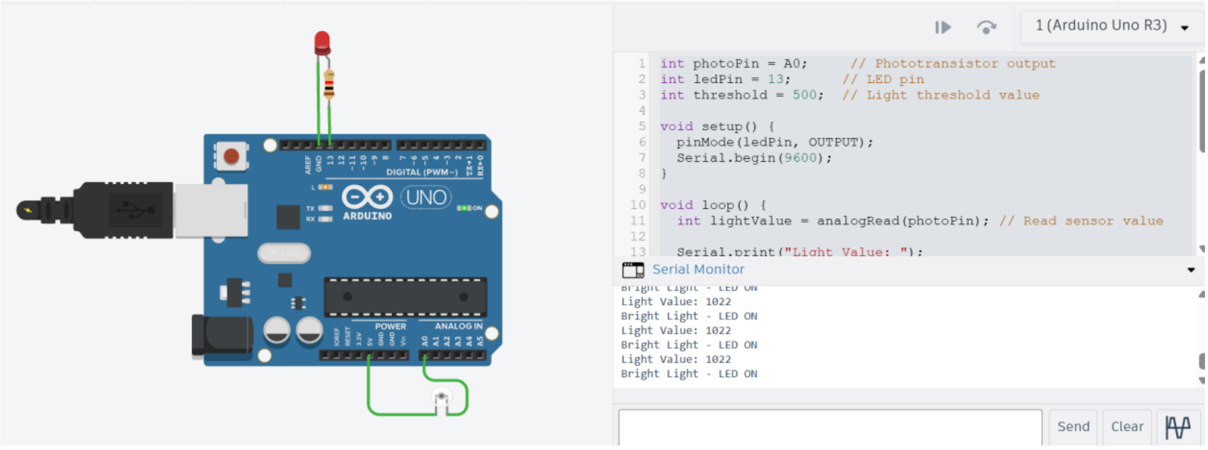
- Reads light intensity using analogRead()
- Displays the value on **Serial Monitor**
- Compares it with a **threshold**
- Turns **LED ON** when light exceeds threshold
- Turns **LED OFF** otherwise

4. Output:-

1) Led OFF:-



2) Led ON:-


1 (Arduino Uno R3)
int photoPin = A0;   // Phototransistor output
int ledPin = 13;   // LED pin
int threshold = 500; // Light threshold value
void setup() {
pinMode(ledPin, OUTPUT);
Serial.begin(9600);
}
void loop() {
int lightValue = analogRead(photoPin); // Read sensor value
Serial Monitor
Light Value: 1022
Bright Light - LED ON
Light Value: 1022
Bright Light - LED ON
Light Value: 1022
Bright Light - LED ON
Send
Clear
ARDUINO
UNO